LAB EXPERIMENT RECORD

Set 15: Product Sales Analysis

Department	AIDS
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
Name	Swetha S
Register Number	192424049
Date	08/10/2026
Day	thursday

AIM

To analyze and visualize first-quarter product sales using R programming and Tableau through grouped bar charts, stacked area charts, and data tables for effective sales comparison and trend analysis.

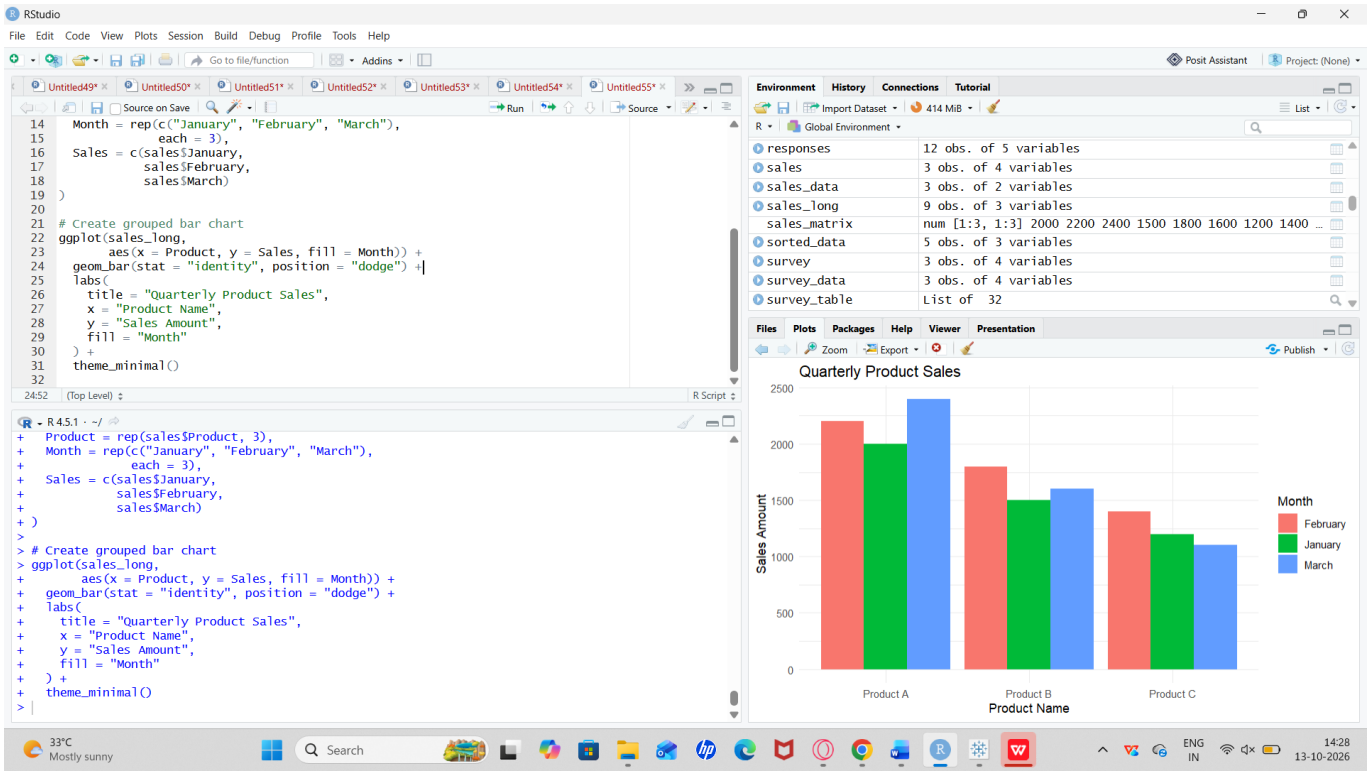
1. Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart elements.

```
library(ggplot2)

sales <- data.frame(
  Product = c("Product A", "Product B", "Product C"),
  January = c(2000, 1500, 1200),
  February = c(2200, 1800, 1400),
  March = c(2400, 1600, 1100)
)

# Convert data into long format
sales_long <- data.frame(
  Product = rep(sales$Product, 3),
  Month = rep(c("January", "February", "March"),
              each = 3),
  Sales = c(sales$January,
            sales$February,
            sales$March)
)

# Create grouped bar chart
ggplot(sales_long,
       aes(x = Product, y = Sales, fill = Month)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(
    title = "Quarterly Product Sales",
    x = "Product Name",
    y = "Sales Amount",
    fill = "Month"
  ) +
  theme_minimal()
```


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled49* Untitled50* Untitled51* Untitled52* Untitled53* Untitled54* Untitled55*
Source on Save Run Source
14 Month = rep(c("January", "February", "March"),
15 each = 3),
16 Sales = c(sales$January,
17 sales$February,
18 sales$March)
19)
20
21 # Create grouped bar chart
22 ggplot(sales_long,
23 aes(x = Product, y = Sales, fill = Month)) +
24 geom_bar(stat = "identity", position = "dodge") +
25 labs(
26 title = "Quarterly Product Sales",
27 x = "Product Name",
28 y = "Sales Amount",
29 fill = "Month"
30) +
31 theme_minimal()
32
24:52 (Top Level) R Script
R 4.5.1 · ~/
+ Product = rep(sales$Product, 3),
+ Month = rep(c("January", "February", "March"),
+ each = 3),
+ Sales = c(sales$January,
+ sales$February,
+ sales$March)
+)
>
> # Create grouped bar chart
> ggplot(sales_long,
+ aes(x = Product, y = Sales, fill = Month)) +
+ geom_bar(stat = "identity", position = "dodge") +
+ labs(
+ title = "Quarterly Product Sales",
+ x = "Product Name",
+ y = "Sales Amount",
+ fill = "Month"
+) +
+ theme_minimal()
>
Environment History Connections Tutorial
Import Dataset 414 MiB List
R Global Environment
responses 12 obs. of 5 variables
sales 3 obs. of 4 variables
sales_data 3 obs. of 2 variables
sales_long 9 obs. of 3 variables
sales_matrix num [1:3, 1:3] 2000 2200 2400 1500 1800 1600 1200 1400 ...
sorted_data 5 obs. of 3 variables
survey 3 obs. of 4 variables
survey_data 3 obs. of 4 variables
survey_table List of 32
Files Plots Packages Help Viewer Presentation
Zoom Export Publish
Quarterly Product Sales
Sales Amount
2500
2000
1500
1000
500
0
Product A
Product B
Product C
Product Name
Month
February
January
March
33°C
Mostly sunny
Search
ENG
IN
14:28
13-10-2026

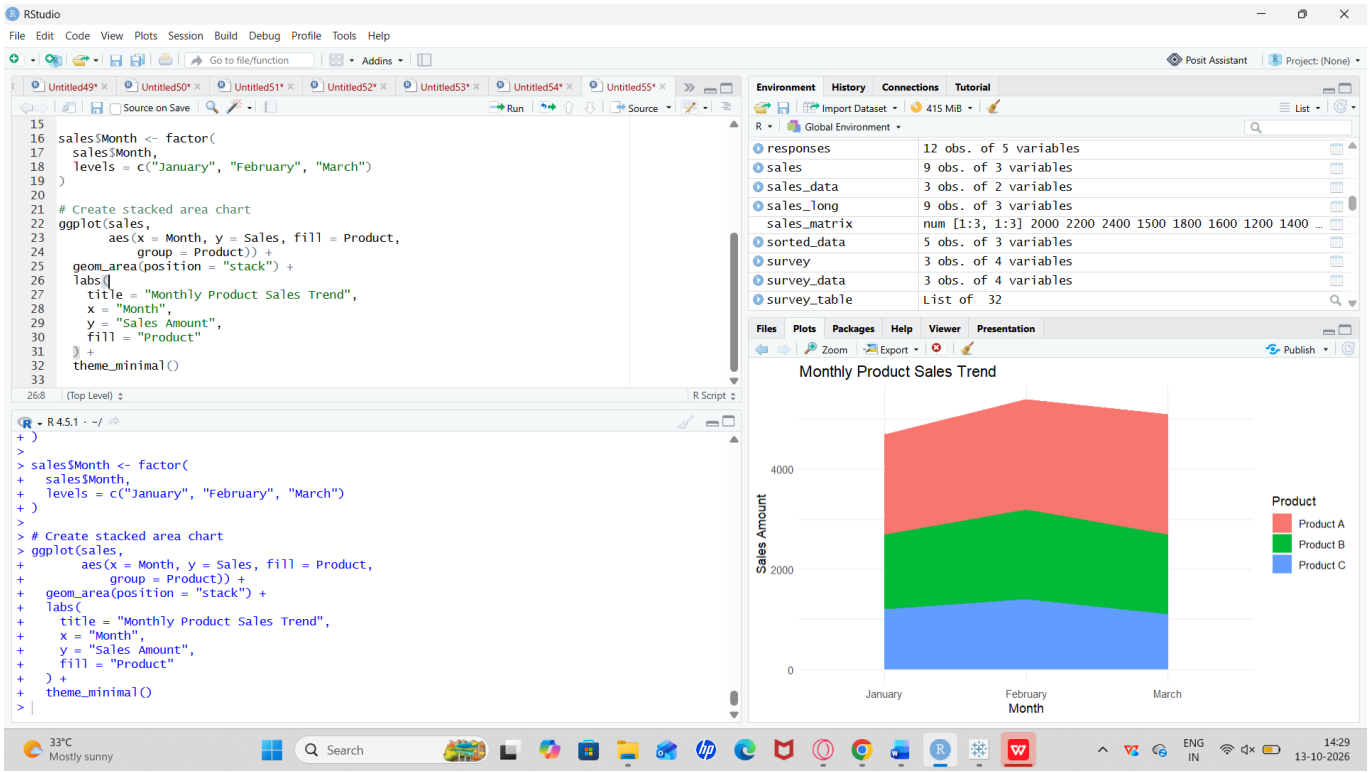
2. Generate a radar chart to visualize the overall pattern of responses across all three questions..

```
library(ggplot2)

sales <- data.frame(
  Month = rep(c("January", "February", "March"),
              each = 3),
  Product = rep(c("Product A", "Product B",
                  "Product C"), 3),
  Sales = c(
    2000, 1500, 1200,
    2200, 1800, 1400,
    2400, 1600, 1100
  )
)

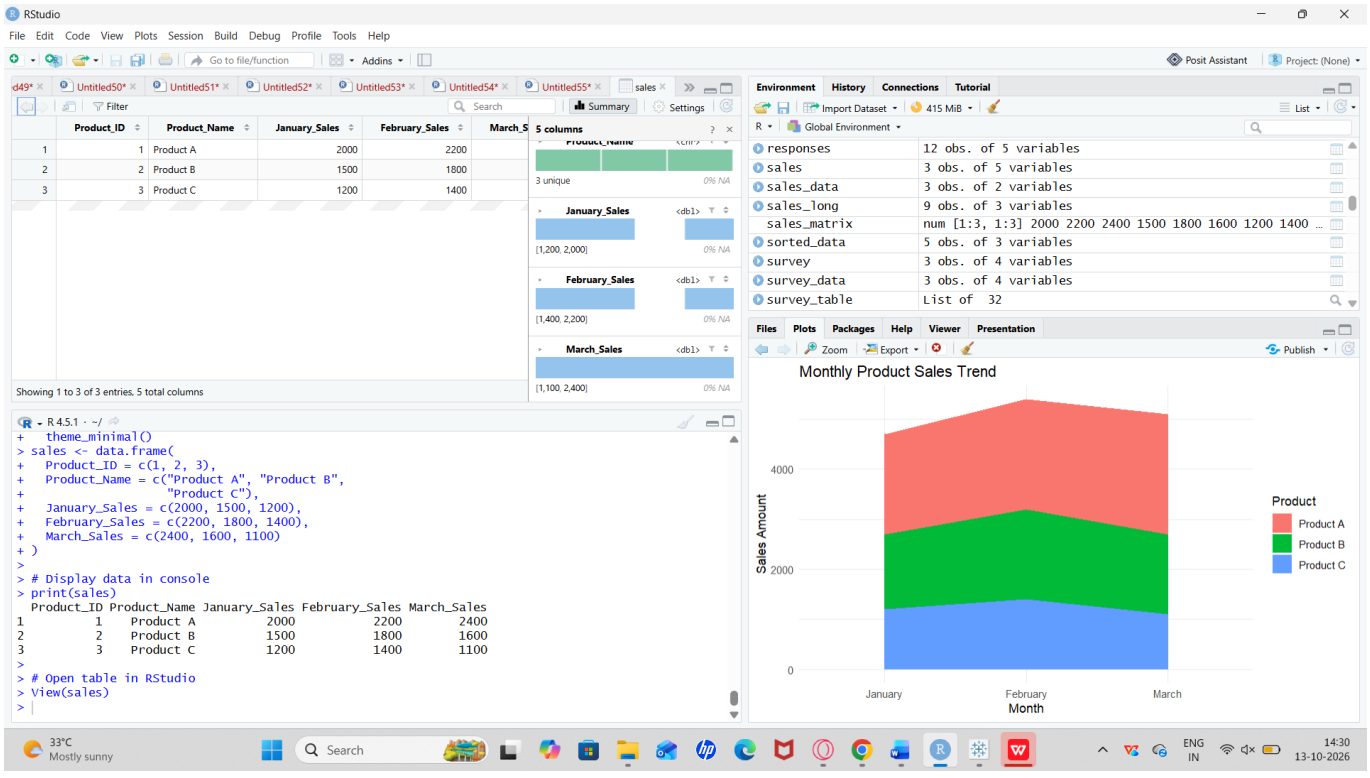
sales$Month <- factor(
  sales$Month,
  levels = c("January", "February", "March")
)

# Create stacked area chart
ggplot(sales,
       aes(x = Month, y = Sales, fill = Product,
           group = Product)) +
  geom_area(position = "stack") +
  labs(
    title = "Monthly Product Sales Trend",
    x = "Month",
    y = "Sales Amount",
    fill = "Product"
  ) +
  theme_minimal()
```

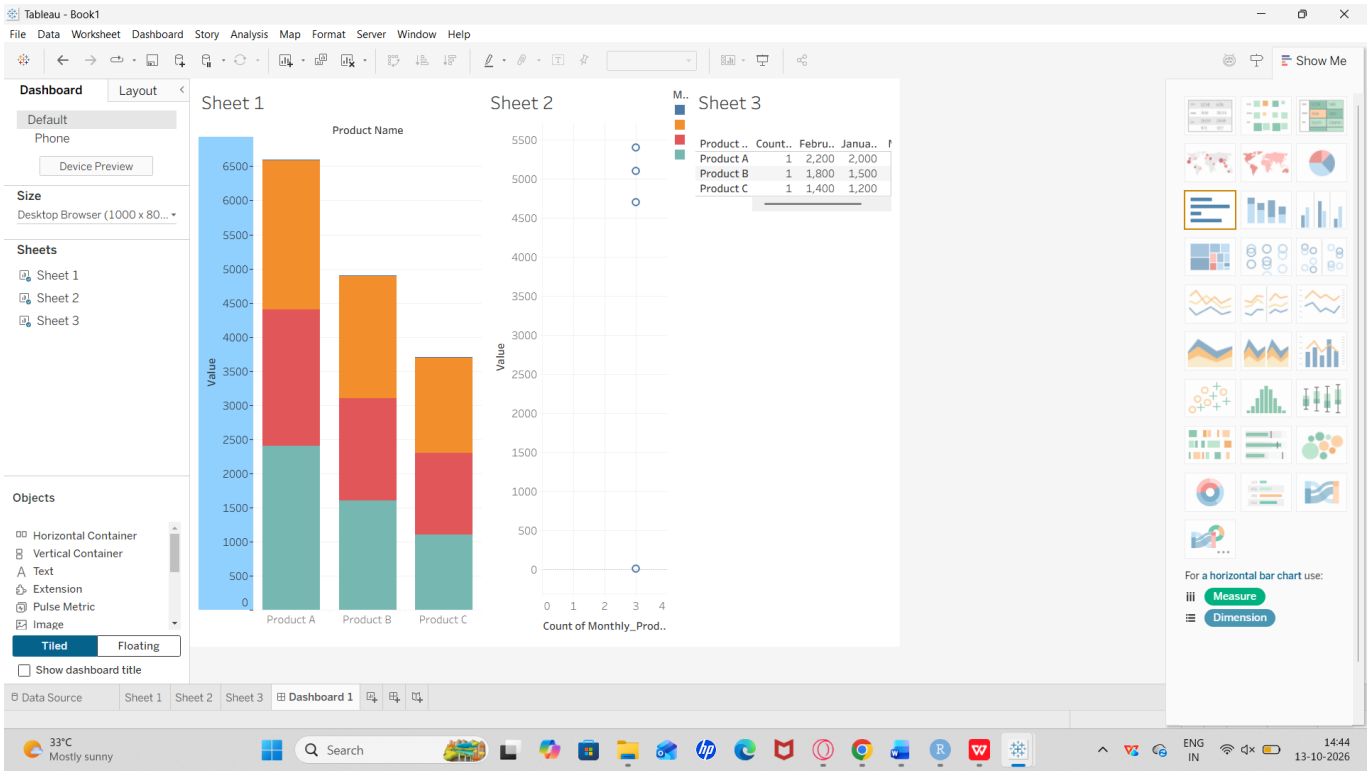

RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled49* Untitled50* Untitled51* Untitled52* Untitled53* Untitled54* Untitled55*
Source on Save Run Source
15
16 sales$Month <- factor(
17 sales$Month,
18 levels = c("January", "February", "March")
19)
20
21 # Create stacked area chart
22 ggplot(sales,
23 aes(x = Month, y = Sales, fill = Product,
24 group = Product)) +
25 geom_area(position = "stack") +
26 labs(
27 title = "Monthly Product Sales Trend",
28 x = "Month",
29 y = "Sales Amount",
30 fill = "Product"
31) +
32 theme_minimal()
33
26:8 (Top Level) R Script
R 4.5.1 · ~/
+)
>
> sales$Month <- factor(
+ sales$Month,
+ levels = c("January", "February", "March")
+)
>
> # Create stacked area chart
> ggplot(sales,
+ aes(x = Month, y = Sales, fill = Product,
+ group = Product)) +
+ geom_area(position = "stack") +
+ labs(
+ title = "Monthly Product Sales Trend",
+ x = "Month",
+ y = "Sales Amount",
+ fill = "Product"
+) +
+ theme_minimal()
>
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Files Plots Packages Help Viewer Presentation
Zoom Export Publish
Monthly Product Sales Trend
Sales Amount
4000
2000
0
January
February
March
Month
Product
Product A
Product B
Product C
33°C
Mostly sunny
Search
ENG
IN
14:29
13-10-2026

3. Build a table to show the survey response data for each respondent. Label the table elements..

```
sales <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B",
                   "Product C"),
  January_Sales = c(2000, 1500, 1200),
  February_Sales = c(2200, 1800, 1400),
  March_Sales = c(2400, 1600, 1100)
)
print(sales)
View(sales)
```



4.



PROCEDURE

1. Create a dataset containing product IDs, product names, and monthly sales.
2. Load the required R package, ggplot2.
3. Create a grouped bar chart to compare monthly sales by product.
4. Generate a stacked area chart to visualize monthly sales trends.
5. Display the product sales data in a table.
6. Export the dataset as a CSV file.
7. Import the CSV file into Tableau and create the required worksheets.
8. Combine the worksheets into an interactive dashboard with filters.

Result

Thus, product sales analysis was performed using R programming by creating a grouped bar chart, stacked area chart, and sales data table. The dataset was exported to CSV format and prepared for integration into Tableau, enabling comparison of product performance and analysis of monthly sales trends.